

to find a cure in the form of vaccinations, experimental treatments or repurposed medication is extremely high, taking precedence over almost everything for the Indian healthcare system and professionals, scientists, startups and researchers. Let's look at some efforts being made in India that are bringing us closer to a cure for COVID-19.

THE RACE FOR A COVID-19 VACCINE

The world is anticipating, with bated breath, a vaccine that can combat the novel coronavirus. There's a massive global effort hurtling towards developing an effective vaccine for COVID-19. The Serum Institute of India and Bharat Biotech are some companies from India that are working on developing a vaccine on a global scale.

Serum Institute of India

Pune-based Serum Institute of India (SII) has joined forces with US-based biotech drug research company, Codagenix, to develop a live-attenuated vaccine (keeping the pathogen alive but less virulent) for COVID-19. The company has stated that the virus-vaccine strain that has been developed in their labs is already in the animal clinical testing stage and the results will arrive within a month or two. The company plans to conduct human clinical trials in India itself and they expect to conduct these trials within the next six months. SII also partnered with Ox-

ford University to manufacture the ChAdOx1 in India. The vaccine was developed in Oxford University and will be produced in India if clinical trials are successful. The ChAdOx1 vaccine utilises a modified adenovirus which carries the coronavirus spike protein that allows it to enter human cells. The cells begin creating and releasing spike proteins and the immune system responds by creating, hopefully, lasting antibodies.

Serum Institute of India's recombinant BCG (Bacillus Calmette-Guerin) vaccine will also undergo human trials to assert whether it boosts immunity in adults against COVID-19. Serum Institute has confirmed that they can produce about 300-400 million doses of this vaccine if the results of the trial are favourable. The vaccine is over 100-years-old with a proven safety record, so it can easily be brought to the market by the end of 2020, according to SII. The BCG vaccine is usually administered to infants at birth in India and globally it is given to children below the age of one to fight tuberculosis. Adar Poonawalla, CEO of SII told Business Standard, "There is a demand for this vaccine, and we can make 300-400 million doses of this vaccine. But, until the trials are over, we are not officially prescribing it." This could be extremely convenient since most other vaccine candidates are yet to reach the human trial phase. Serum Institute will most likely begin trials in Maharashtra with 2,000-3,000 high-risk infected patients. Some recent studies also indicate that countries that do not administer a universal BCG vaccination have reported higher instances of COVID-19 cases in comparison to countries that do.

Bharat Biotech

Bharat Biotech, a biopharmaceutical firm in India, has partnered with the University of Wisconsin-Madison as well as US-based FluGen to develop an intranasal vaccine, which they have dubbed 'CoroFlu'. This vaccine

will build on the backbone of FluGen's M2SR flu vaccine. The FluGen lab will insert gene sequences from SARS-CoV-2 (the novel coronavirus that causes COVID-19) into M2SR. This is done so that the new vaccine provides immunity against COVID-19 as well as influenza. FluGen co-founders, Bharat Biotech mentioned in a release that 'CoroFlu' stands apart from other competing vaccines in development since its principle is already proven to be safe and well-tolerated in humans in Phase I and II clinical trials.



Bharat Biotech is working on an intranasal vaccine called 'CoroFlu'

The Head of Business Development at Bharat Biotech, Raches Ella stated that the company intends to manufacture the vaccine, conduct clinical trials, and then prepare to mass-produce around 300 million doses for distribution across the globe. FluGen has a collaboration agreement with Bharat Biotech, where it will transfer the manufacturing process to the Indian company in order to scale production. Estimates say that the CoroFlu vaccine, now undergoing animal testing in the lab at UW-Madison, will likely be ready for human trials in three to six months. Bharat Biotech has commercialised 16 vaccines including one for the H1N1 flu in 2009.

Formation of a task force to develop a vaccine

India has also formed a high-level task force dedicated to conducting research on the novel coronavirus and developing a vaccine for the same. The task force will be co-chaired by Professor K Vijay Raghavan, Principal Scientific Adviser to the Govern-



The BCG vaccine could be repurposed for COVID-19